

Preparing Job Scripts (sbatch)

#SBATCH Directives

Applications have to be run in batch mode on the supercomputer. This means that a job script must be prepared for each run, which contains the description of the required resources and the commands required for the run. The script can be submitted with the `sbatch` command to the Slurm scheduler. Parameters for the scheduler (resource requirements) can be provided at the top of the script using the `#SBATCH` directive in the script. (Note, that sbatch will stop processing further sbatch directives once the first non-comment non-whitespace line has been reached in the script.)

Basic Options 🔗

The most basic options that can be provided using sbatch directives are demonstrated in the following sbatch script:

```
#!/bin/bash
#SBATCH --account=ACCOUNT
#SBATCH --job-name=NAME
#SBATCH --partition=PARTITION
#SBATCH --time=TIME
#SBATCH --cpus-per-task=NCPUS
#SBATCH --mem-per-cpu=SIZE

EXECUTABLE COMMANDS
...
```

📌 Important

If an option is not set, Slurm applies the default value for that option. You should always make sure that either you set the options correctly or that the default is suitable for your job, otherwise your job may not run as expected (e.g. may be prioritized low, runs out of memory, will not be able to scale up properly or even may be pending forever).

Description of the options above (and their defaults):

`--account (-A)`: *ACCOUNT* is the name of the project account to be debited (accessible accounts can be listed using the `sbalance` command). **Default:** The account (project) associated with the owner of the job (on Komondor, each user is associated with exactly one account for each project in which the user is participating).

`--job-name (-J)`: *NAME* is the short name of the job. **Default:** The name of the batch script or the sender application.

`--partition (-p)`: *PARTITION* is the partition that is requested for the resource allocation. **Default:** `cpu`

`--time (-t)`: *TIME* is the maximum running time (walltime) allowed for the job. The following time formats can be used: "MINUTES", "MINUTES:SECONDS", "HOURS:MINUTES:SECONDS", "DAYS-HOURS", "DAYS-HOURS:MINUTES" and "DAYS-HOURS:MINUTES:SECONDS". On the main compute partitions (`cpu`, `gpu`, `ai`, `bigdata`) the **default is 2 days** and the **maximum is 7 days**. Some specialized partitions have shorter limits (e.g. `test`: default 10 minutes, max 1 hour; `cpu-short`: default 1 hour, max 2 hours).

`--cpus-per-task (-c)`: *NCPUS* is the number of processors that is to be allocated per task within the job. The **default is one CPU core per task**.

`--mem-per-cpu`: *SIZE* is the minimum memory required per usable allocated CPU. Default units are **megabytes**. On the Komondor, the **default is 1000 MB / CPU core**. The **maximum** memory that can be allocated **per CPU core** varies per partition: "CPU" - 2000 MB, "GPU" - 4000 MB, "AI" - 4000 MB, "BigData" - 42000 MB, "test" - 4000 MB, "cpu-short" - 4000 MB. (if more memory allocation is specified, a job will be billed for the number of CPUs which can provide the required amount of memory)

For a full list of sbatch options, see the [description of sbatch in the Slurm documentation](#) or type `man sbatch` in the terminal.

Time Limit

Running jobs have a time limit for which they are allowed to run. When the time limit expires, the job is canceled by the scheduler. The default time limit on Komondor is **2 days**. You can explicitly set the time limit for your job with the `--time` option. The

maximum time limit that can be requested is **7 days** on the main compute partitions (`cpu`, `gpu`, `ai`, `bigdata`). Some specialized partitions (such as `test` and `cpu-short`) enforce shorter limits.

```
#SBATCH --time=TIME
```

TIME is the maximum running time (walltime) allowed for the job. The following time formats can be used: “MINUTES”, “MINUTES:SECONDS”, “HOURS:MINUTES:SECONDS”, “DAYS-HOURS”, “DAYS-HOURS:MINUTES” and “DAYS-HOURS:MINUTES:SECONDS”.

The current time limits set for the partitions can be queried with the following command:

```
sinfo --Format partition,defaulttime,time
```

Note

Shorter jobs can get higher priority, so it's a good idea to set the job's time limit as accurate as you can estimate. Properly setting the time limit also helps Slurm to schedule jobs more efficiently.

CPU Allocation

By default, one CPU core is allocated per task within your job. You can ask for more processors with the `--cpus-per-task` option:

```
#SBATCH --cpus-per-task=NCPUS
```

NCPUS is the number of processors that is to be allocated per task within the job.

Memory Allocation

By default, 1000 MB of memory is assigned to 1 allocated CPU core. You can request more with the `--mem-per-cpu` option:

```
#SBATCH --mem-per-cpu=SIZE
```

SIZE is given in MB (megabytes) by default. You can specify the unit explicitly: K for KB (kilobytes), M for MB (megabytes), G for GB (gigabytes), T for TB (terabytes).

The maximum memory that can be allocated per CPU core varies per partition:

- “CPU”: 2000 MB / core
- “GPU”: 4000 MB / core
- “AI”: 4000 MB / core
- “BigData”: 42000 MB / core
- “test”: 4000 MB / core
- “cpu-short”: 4000 MB / core

You can use the `--mem-per-gpu` option to specify the amount of memory required per allocated GPU:

```
#SBATCH --mem-per-gpu=SIZE
```

The `--mem` option sets the required memory per allocated node:

```
#SBATCH --mem=SIZE
```

Here, setting the *SIZE* to 0 means all of the memory on the allocated nodes.

Note: The `--mem-per-cpu`, `--mem-per-gpu` and `--mem` options are mutually exclusive.

GPU Allocation

GPUs can be allocated using the `--gres` option:

```
#SBATCH --gres=gpu:N
```

N sets the number of required GPUs per node, which can be 1-4 (“GPU” partition) or 1-8 on the “AI” partition (nodes on “CPU” and “BigData” partitions don’t have GPUs).

Node Allocation

Slurm will allocate enough nodes to satisfy the requested resources. However, you can explicitly specify the number of nodes you want to assign to your job with the `--nodes` option:

```
#SBATCH --nodes=N
```

N sets the number of required nodes for the job. You can also set this option as e.g. “MINIMUM-MAXIMUM” or a comma-separated list of node numbers.

Setting the number of required nodes does not mean you will get all resources of the allocated nodes. It just means that the tasks of your job can use (can be distributed over) that many nodes. If you want to allocate all the CPUS and GRES (generic resources, e.g. GPUs) on the nodes for your job, you can use the `--exclusive` option. Note, that this does not mean that you also get all the memory on the allocated nodes.

```
#SBATCH --nodes=N
#SBATCH --exclusive
#SBATCH --mem=SIZE_PER_NODE
```

Non-restartable Jobs

In case of jobs that are not restartable or should not restart, the `--no-requeue` option can be set to prevent requeuing (e.g. after node failure):

```
#SBATCH --no-requeue
```

This setting needs to be applied since default behaviour of Komondor is to requeue the jobs.

Quality of Service

Each job submitted to Slurm is assigned a Quality of Service (QoS), that affects the execution (e.g. priority, preemption, interruption, resource billing) of the job. The default QOS is “normal”: the job cannot be interrupted, and as much CPU time is billed

as was used.

For more detailed description and information about the available QOSs, see the [“Basic Usage / Quality of Service \(QOS\)”](#) chapter.

You can set a QOS other than the default for your job using the `--qos` option, for example, here is how you set the QOS to “lowpri”:

```
#SBATCH --qos=lowpri
```

Low priority jobs may be interrupted then resumed at any time. It is advised to test the behaviour of the used software by artificially terminating it (more explanation will follow...).

Email Notification

You can instruct Slurm to send an email when the state of your job changes (e.g. starts, ends or gives an error):

```
#SBATCH --mail-type=ALL
#SBATCH --mail-user=EMAIL
```

You can set the triggering events using the `--mail-type` option (you can find the full list in the [sbatch description in the Slurm documentation](#) or type `man sbatch` on the terminal). The `--mail-user` option sets the email address where the notification should be sent.

Slurm Environment Variables

Many of the configurations specified (explicitly or implicitly) in the sbatch script are accessible with environmental variables after submission. Some examples:

<code>\$SLURM_JOB_ID</code>	Unique ID for the job.
<code>\$SLURM_JOB_NAME</code>	Name of your job (set with <code>--job-name</code>).
<code>\$SLURM_CPUS_PER_TASK</code>	Number of CPU cores per task (set with <code>--cpus-per-task</code>).
<code>\$SLURM_MEM_PER_CPU</code>	Amount of memory per CPU core (set with <code>--mem-per-cpu</code>).

<code>\$SLURM_MEM_PER_NODE</code>	Amount of memory per node (set with <code>--mem</code>).
<code>\$SLURM_NTASKS</code>	Number of tasks (set with <code>--ntasks</code>).
<code>\$SLURM_NTASKS_PER_NODE</code>	Number of tasks per node (set with <code>--ntasks-per-node</code>).

You can find a [full list of Slurm output variables in the Slurm documentation](#) 